Proceedings of the ASME/BATH 2014 Symposium on Fluid Power & Motion Control
FPMC2014
September 10-12, Bath, United Kingdom

FPMC2014-7857

EXPERIMENTAL VALIDATION OF A SOFT SWITCH FOR A VIRTUALLY VARIABLE DISPLACEMENT PUMP

Brandon K. Beckstrand, M.S. Candidate
University of Minnesota
Minneapolis, MN, USA

Prof. James D. Van de Ven
University of Minnesota
Minneapolis, MN, USA

ABSTRACT

As an alternative to a variable displacement pump, a fixed displacement pump can be made to function as a virtually variable displacement unit by using a high-speed valve to pulse-width modulate the flow, creating a switch-mode circuit. A major drawback of switch-mode circuits is throttling and compressibility energy losses during valve transitions. One method of minimizing these losses is soft switching, where the flow that would normally be throttled across the high speed valve during transitions is absorbed in a small variable volume chamber. The concept for a novel soft switch mechanism that uses the pressure signal at the exit of the pump to release a lock on the soft switch chamber was previously presented. This paper describes the soft switch concept in more detail and outlines a numerical model used to predict and optimize soft switch operation. Experimental results are presented that demonstrate proper operation of the soft switch lock-release mechanism.

INTRODUCTION/BACKGROUND

Common methods of controlling the flow rate in hydraulic power systems include variable displacement pumps as well as the use of metering valves. These control methods are inefficient in the case of metering valves, and bulky and expensive in the case of variable displacement pumps [1]. Another variable displacement control scheme uses the concept of pulse-width modulation, which is comparable to switch mode control in electronic power circuits [2]. In hydraulic switch mode circuits, a high speed valve quickly switches back and forth between on and off positions, controlling the average pressure and flow rate to the load line of the circuit. These digital hydraulic circuits are theoretically more efficient and compact than throttling valve control [1].

Unfortunately, switch mode hydraulic circuits are not often used because their efficiency is decreased in practice by a few energy loss mechanisms. The main losses, as stated by Cao et al., are internal leakage, viscous friction, throttling during valve transitions, inertial forces of the high speed valve, and fluid compressibility [3]. Of those five loss mechanisms throttling across the high speed valve accounts for 15 – 60% of the total energy losses in the system depending on circuit and valve designs [4,5]. Rannow and Li proposed using a soft switch to absorb the flow that would normally be throttled across the partially opened high speed valve during valve transitions [6]. The soft switching approach helps address the throttling and compressibility losses since the flow rate across the valve during the transitions is lowered, which results in less fluid compression and decompression in the switched volume.

The soft switch concept as described by Rannow and Li is a small spring-loaded piston cylinder device that has an externally actuated lock at an intermediate location of the piston travel. One major challenge that has made the soft switch concept difficult to implement is the need for precise external electromechanical locking/unlocking components. A novel soft switch was proposed by Van de Ven that solves the problem of precise timing of the piston release by controlling the unlocking of the piston via the balance of pressure forces on the front and back of the piston [7].

The first two sections of this paper describe the basic operation of the soft switch and its associated locking mechanism. The next section outlines the design and testing of the physical system with emphasis on parameters that allow for easy cross-test comparison. The last section provides details on actual soft switch operation and system efficiency.

NOMENCLATURE

$A_{ch,n}$	*Check valve area of entity n [m²]*
A_{lock}	*Area of locking port to tank [m²]*
A_s	*Cross-sectional area of the soft switch piston [m²]*
A_n	*3-way valve port area for entity n [m²]*
C_d	*Discharge coefficient [-]*

Copyright © 2014 by ASME

Downloaded From: http://proceedings.asmedigitalcollection.asme.org/ on 01/29/2016 Terms of Use: http://www.asme.org/about-asme/terms-of-use

c	*Clearance between soft switch piston and cylinder [m]*
d	*Diameter of small port in loss calculations [m]*
d_s	*Diameter of the soft switch piston [m]*
$d_{ch,n}$	*Diameter of the check valve poppet entity n [m]*
D	*Diameter of large port in loss calculations [m]*
F_{prel}	*Preload of soft switch spring [N]*
k	*Loss coefficient [-]*
k_s	*Rate of the soft switch spring [N/m]*
$k_{ch,n}$	*Rate of the check valve spring for entity n [N/m]*
$l_{back,a}$	*Length of leakage path from piston annulus to cylinder tank port [m]*
$l_{back,b}$	*Length of leakage path from back of soft switch to cylinder tank port [m]*
$l_{ch,n}$	*Length of the check valve poppet for entity n [m]*
l_{front}	*Length of leakage path from front of soft switch to cylinder tank port [m]*
l_s	*Length of the piston in contact with the cylinder [m]*
m_s	*Mass of the piston [kg]*
$m_{ch,n}$	*Mass of check valve entity n [kg]*
P_{back}	*Pressure behind the soft switch [Pa]*
$P_{crk,n}$	*Cracking pressure of check valve entity n [Pa]*
P_{load}	*Pressure of the load line [Pa]*
P_{switch}	*Pressure in the switched volume [Pa]*
P_{tank}	*Tank pressure [Pa]*
P_0	*Atmospheric pressure [Pa]*
$Q_{ch,n}$	*Flow rate through check valve entity n [m³/s]*
$Q_{leak,b}$	*Leakage flow rate from back of the soft switch to locking annulus [m³/s]*
$Q_{leak,f}$	*Leakage flow rate from front of the soft switch to locking annulus [m³/s]*
Q_{lock}	*Flow rate through the locking port [m³/s]*
$Q_{lock,loss}$	*Flow rate lost through locking port [m³/s]*
Q_{pump}	*Flow rate of the pump [m³/s]*
$Q_{HSV,n}$	*Flow rate through the 3-way valve to entity n [m³/s]*
R	*Volume fraction of air entrained in the oil [-]*
V_{back}	*Volume behind the soft switch piston [m³]*
V_{dead}	*Dead volume behind the soft switch piston [m³]*
v_s	*Velocity of the soft switch [m/s]*
$v_{ch,n}$	*Velocity of check valve entity n [m/s]*
x_s	*Position of the soft switch [m]*
$x_{ch,n}$	*Position of check valve entity n [m]*
α	*Kinetic energy correction factor [-]*
β	*Bulk modulus of air free oil [-]*
β_e	*Effective bulk modulus [-]*
μ	*Dynamic viscosity [Pa*s]*
γ	*Ratio of specific heats for air [-]*
ρ	*Mass density of oil [kg/m³]*

Subscripts

l	*Load*
t	*Tank*
HSV	*3-way valve (high speed valve)*

CONCEPT

The concept as proposed by Van de Ven [7] will now be outlined, including a description normal operation. Fig. 1 shows the proposed soft switch mechanism. The piston has an internal port that connects the right side of the piston to the annulus in the middle of the piston. It is seated in the left end of the cylinder via a compression spring on its right side. The cylinder has an annular port that transitions to circular ports that align with the piston annulus as the piston is displaced to the right.

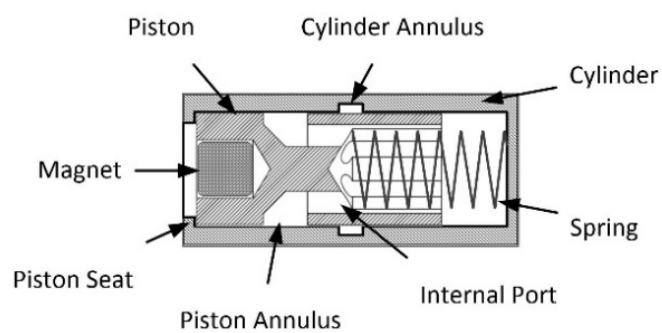

Figure 1 - Soft switch mechanism.

In order to understand the locking mechanism operation, consider a virtually variable displacement pumping circuit that utilizes a high speed 3-way valve and the soft switch, as shown in Fig. 2. The left or front side of the soft switch cylinder is connected to the switched volume, while the annular cylinder port is connected to tank, and the right or back side of the soft switch cylinder is connected to tank and the high pressure rail via check valves. At the beginning of the switching cycle the 3-way valve is in the tank position and the soft switch is empty with the piston seated to the left. When the 3-way valve is transitioning from tank to load, the pressure in the switched volume begins to rise, causing the piston to move slightly to the right and compress the fluid on the right side of the soft switch piston. When the 3-way valve completes its transition to load, the piston remains stationary, other than movement due to leakage. As the pressure port begins to close, the pressure in the switched volume exceeds the high pressure rail, causing the pressure check valve to open and the soft switch piston to displace. When the piston has displaced to the point where the annulus on the piston aligns with the annulus on the cylinder, a

Copyright © 2014 by ASME

Downloaded From: http://proceedings.asmedigitalcollection.asme.org/ on 01/29/2016 Terms of Use: http://www.asme.org/about-asme/terms-of-use

sudden decrease in pressure on the back side of the piston occurs, unlocking the piston. The soft switch then rapidly absorbs flow at tank pressure, which minimizes the throttling losses that would normally be incurred as the 3-way valve transitions back from load to tank. Once the 3-way valve has completely opened to tank, the spring forces the piston back to the left, which sends the fluid inside the soft switch to tank, while fluid enters the back side of the soft switch from tank. Now that the piston is reseated, the next switching cycle can begin.

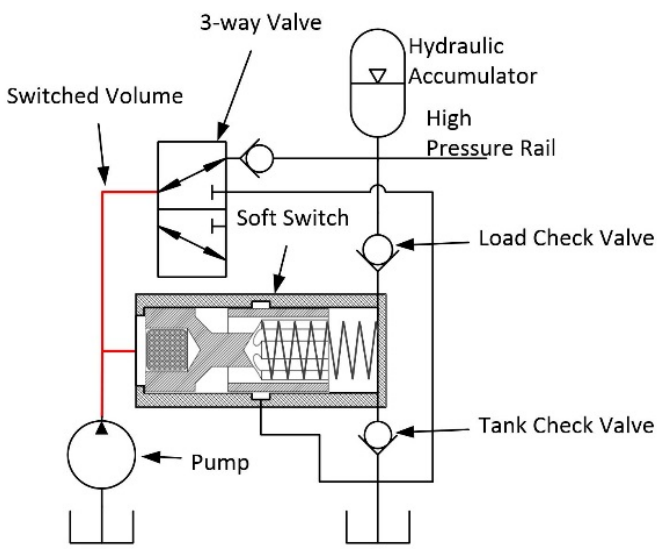

Figure 2 - Virtually variable displacement pumping circuit with soft switch.

Rannow and Li originally proposed a two-state soft switch that absorbed flow during both transitions [6]. Van de Ven expanded upon this by proposing two soft switch chambers, one passive and the other active, to absorb both 3-way valve transitions separately [7]. The two switch chambers would be placed in parallel in the switched volume line with the passive switch optimized to absorb the flow during transitions to load, and the active switch unlocking and absorbing flow during the transitions back to tank. Though the dual soft switch approach has been shown in simulations to offer higher efficiencies, since the purpose of this work is to show the feasibility of the unlocking mechanism it was not implemented in this work.

NUMERICAL MODEL

A dynamic model of the soft switch with proposed release mechanism was given by Van de Ven [7]. This section reviews and builds upon the model previously presented. The pump is assumed to be a constant flow source and the accumulator allows the load line to maintain constant pressure. All fluid conductors between components are assumed to behave ideally, with resistances and inertia being negligible.

The pressures in the switched volume and behind the soft switch are derived using the definition of the fluid bulk modulus, $\beta_e = -V(dP/dV)$, with the switched volume pressure shown in Eqn. (1) and the pressure behind the soft switch shown in Eqn. (2).

$$\dot{P}_{switch} = \frac{\beta_e(P_{switch})}{(A_s x_s + V_{switch})} \times (Q_{pump} - Q_{HSV,l} - Q_{HSV,t} - Q_{leak,f} - A_s v_s) \tag{1}$$

where Q_{pump} is the flow rate out of the pump, $Q_{HSV,l}$ and $Q_{HSV,t}$ are the flow rates through the 3-way valve to load and tank respectively, $Q_{leak,f}$ is the leakage flow rate from the switched volume to the locking port, A_s is the cross-sectional area of the piston, v_s and x_s are the velocity and position of the piston respectively, and V_{switch} is the switched volume.

$$\dot{P}_{back} = \frac{\beta_e(P_{back})}{(V_{back} + V_{dead})} \times (Q_{ch,t} + Q_{leak,f} - Q_{ch,l} - Q_{lock} - Q_{leak,b} + A_s v_s) \tag{2}$$

where $Q_{ch,t}$ and $Q_{ch,l}$ are the flow rates through the tank and load check valves respectively, Q_{lock} is the flow rate through the locking port, $Q_{leak,b}$ is the leakage flow rate from the back of the soft switch to the locking port, V_{back} is the volume behind the piston, and V_{dead} is the dead volume behind the piston.

Notice that though leakage past the piston is considered, the leakage flow past the high speed valve is considered to be negligible. The effective bulk modulus is calculated using Eqn. (3), developed by Cho et al. [8].

$$\beta_e(P) = \beta \left[\frac{\left(\frac{P}{P_0}\right)^{\frac{1}{\gamma}} e^{\frac{(P_0 - P)}{\beta}} + R}{\frac{R}{\gamma}\frac{\beta}{P} + \left(\frac{P}{P_0}\right)^{\frac{1}{\gamma}} e^{\frac{(P_0 - P)}{\beta}}} \right] \tag{3}$$

where β is the bulk modulus of air free oil, P_0 is the atmospheric pressure, γ is the ratio of specific heats for air, R is the volume fraction of air entrained in the oil.

The flow rate through the 3-way valve is modeled using the orifice equation with an open area that varies with time. If the valve is switched to tank, the flow rate is described as:

$$Q_{HSV,t} = C_d A_t \sqrt{2/\rho |P_{switch} - P_{tank}|} \times \operatorname{sgn}(P_{switch} - P_{tank}) \tag{4}$$

where C_d is the discharge coefficient, A_t is the 3-way valve tank port area, plotted as a function of time in Fig. Figure 3, ρ is the mass density of oil, and P_{tank} is the tank pressure.

If the valve is switched to load, then the flow goes through the 3-way valve and a check valve, so the flow is affected by both in series. The check valve is assumed to open and close instantaneously when the difference in pressures across it reaches the cracking pressure. As a result of the check valve, flow is only allowed through the 3-way valve when the check valve is cracked. This flow rate is described as:

Copyright © 2014 by ASME

Downloaded From: http://proceedings.asmedigitalcollection.asme.org/ on 01/29/2016 Terms of Use: http://www.asme.org/about-asme/terms-of-use

$$\text{if } P_{switch} - P_{load} > P_{crk,HSV}$$
$$Q_{HSV,l} = \frac{{C_d}^2 A_l A_{ch,HSV}}{\sqrt{{C_d}^2 {A_l}^2 + {C_d}^2 {A_{ch,HSV}}^2}} \times \sqrt{2/\rho\,(P_{switch} - P_{load} - P_{crk,HSV})} \quad (5)$$
$$\text{else } Q_{HSV,l} = 0$$

where P_{load} is the load line pressure, which is assumed constant, $P_{crk,HSV}$ is the cracking pressure of the check valve in line with the 3-way valve, A_l is the 3-way valve load port area, plotted as a function of time in Fig. 3, and $A_{ch,HSV}$ is the area of the check valve in line with the 3-way valve.

The 3-way valve has a circular tank port, and a rectangular pressure port, which were measured to determine the maximum open area at the extreme displacement conditions. Though the tank port is circular, the valve spool does not displace enough in that extreme position to fully open the port, effectively changing the port area to a partial circular opening. It is also assumed that the spool transitions at a constant velocity. During the spool transition there is also a 0.127 mm travel length where neither port is open. Note also that the actual duty ratio of the valve is less than the command signal (60% duty in this case) because the delay between the time the solenoid valve transition signal is sent and the valve begins to transition to load is larger than the delay for the valve transition to tank. Fig. Figure 3 incorporates an approximate 3-way valve delay based on the delay seen in experimental data.

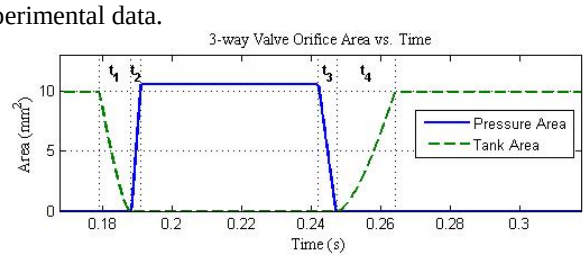

Figure 3 - 3-way valve orifice area as a function of time, with a 60% duty ratio and a period of 0.15 seconds.

The flow rates through the check valves (Eqns. (6) and (7)) connected to the back of the soft switch are calculated using the orifice equation, and the check valves are assumed to have no leakage flow when they are fully seated.

$$Q_{check,l} = C_d A_{ch,l} \sqrt{2/\rho\,|P_{back} - P_{load} - P_{crk,l}|} \times \operatorname{sgn}(P_{back} - P_{load} - P_{crk,l}) \quad (6)$$

$$Q_{check,t} = C_d A_{ch,t} \sqrt{2/\rho\,|P_{tank} - P_{back} - P_{crk,t}|} \times \operatorname{sgn}(P_{tank} - P_{back} - P_{crk,t}) \quad (7)$$

where $A_{ch,l}$ and $P_{crk,l}$ are the area and cracking pressure respectively of the load check valve, $A_{ch,t}$ and $P_{crk,t}$ are the area and cracking pressure respectively of the load check valve.

The flow through the locking port to tank is also modeled using the orifice equation (Eqn. (8) when the open area is greater than zero.

$$Q_{lock} = C_d A_{lock} \sqrt{2/\rho\,|P_{back} - P_{tank}|} \times \operatorname{sgn}(P_{back} - P_{tank}) \quad (8)$$

where A_{lock} is the area of the locking port. The locking port area depends on the location of the soft switch piston. When the piston reaches the unlocking distance it begins to simultaneously open up to 6 circular ports to tank. When the piston reaches its extreme displacement condition the ports to tank have begun to be blocked again, rendering their profiles part circles. The diameter of the circular port is 5.08 mm, so the fully open port area is 121.6 mm^2.

Leakage flow past the soft switch piston can be described by assuming laminar flow between parallel plates. There are two leakage paths between the piston and cylinder. The first is from the back of the soft switch to the unlocking port on the cylinder, which can be divided into two sections. The first section flows from the right side of the piston to the tank port, while the other flows from the unlocking annulus of the piston to the tank port. Combining those flows gives the total leakage from the back of the soft switch to the unlocking tank port (Eqn. (9)). The other leakage flow (Eqn. (10)) goes from the front of the piston to the unlocking annulus of the piston.

$$Q_{leak,b} = \frac{(P_{back} - P_{tank}) 2\pi d_s c^3}{3\mu} \times \left(\frac{1}{l_{back,a}} + \frac{1}{l_{back,b}}\right) \quad (9)$$

$$Q_{leak,f} = \frac{(P_{switch} - P_{back}) 2\pi d_s c^3}{3\mu l_{front}} \quad (10)$$

where d_s is the diameter of the piston, c is the clearance between the piston and cylinder, μ is the dynamic viscosity of the fluid, $l_{back,a}$, $l_{back,b}$, and l_{front} are the lengths of the leakage paths from the piston annulus, the back of the piston, and the front of the piston to the cylinder tank port respectively.

The velocity (Eqn. (11)) and position (Eqn. (12)) of the piston are found using Newton's second law, considering viscous shear of the fluid as a damping term. The displacement of the piston is limited to a range from 0 to $x_{s,max}$ via conditional statements.

$$\dot{v}_s = \frac{(P_{switch} - P_{back}) A_s}{m_s} - \frac{F_{prel} - k_s x_s - \frac{\mu v_s \pi d_s l_s}{c}}{m_s} \quad (11)$$

Copyright © 2014 by ASME

Downloaded From: http://proceedings.asmedigitalcollection.asme.org/ on 01/29/2016 Terms of Use: http://www.asme.org/about-asme/terms-of-use

$$\dot{x}_s = v_s \tag{12}$$

where F_{prel} and k_s are the preload and rate of the spring respectively, l_s is the length of the piston in contact with the cylinder, and m_s is the mass of the piston.

The check valves ported to the volume behind the soft switch piston are poppet style and have velocities (Eqns. 13a and 13b) and positions (Eqns. 14a and 14b) calculated in the same manner as the piston, with displacement limits given by conditional statements from 0 to $x_{l,max}$ and 0 to $x_{t,max}$.

$$\dot{v}_{ch,l} = \frac{(P_{back} - P_{load})A_{ch,l}}{m_{ch,l}} - \frac{P_{crk,l}A_{ch,l} - k_{ch,l}x_{ch,l} - \frac{\mu v_{ch,l}\pi d_{ch,l}l_{ch,l}}{c_{ch,l}}}{m_{ch,l}} \tag{13a}$$

$$\dot{v}_{ch,t} = \frac{(P_{tank} - P_{back})A_{ch,t}}{m_{ch,t}} - \frac{P_{crk,t}A_{ch,t} - k_{ch,t}x_{ch,t} - \frac{\mu v_{ch,t}\pi d_{ch,t}l_{ch,t}}{c_{ch,t}}}{m_{ch,t}} \tag{13b}$$

$$\dot{x}_{ch,l} = v_{ch,l} \tag{14a}$$

$$\dot{x}_{ch,t} = v_{ch,t} \tag{14b}$$

where k_{ch}, x_{ch}, v_{ch}, d_{ch}, l_{ch}, c_{ch}, and m_{ch} are the spring rate, displacement, velocity, diameter of the poppet, thickness of the poppet, clearance between the poppet and body, and mass of the poppet respectively for both load and tank check valves.

The open areas of the check valves behind the soft switch are cylindrical in shape, with length given by the displacement between the poppet and its respective seat, and circumference given by the circumference of the poppet.

PHYSICAL DESIGN AND EXPERIMENTAL METHODS

The following section briefly presents an optimization that was done for the soft switch parameters, followed by a presentation of the physical implementation of the soft switch. The methods for validating the functionality of the soft switch locking mechanism are then described.

The numerical model described by Van de Ven [7], which is similar to the one described in this work, was used to optimize the design of a soft switch for a similar circuit architecture. Van de Ven [7] used a grid search method with the objective being to minimize the energy loss per second of operation. The energy loss per second of operation is calculated by subtracting the energy going to load from the energy delivered from the pump. The parameters that were varied were piston radius, maximum piston displacement, preload of the compression spring, and unlocking distance, with check valve parameters, switching frequency, and duty ratio being held constant. The optimized parameters shown in Table 1 were used as a starting point in determining dimensions for a proof of concept soft switch.

Table 1 - Optimized soft switch parameters.

Parameter	Units	Value
Piston radius	mm	4.5
Max travel of piston	mm	11
Spring preload	N	17.5
Unlocking distance	mm	3.6
Volume behind piston	cm^3	1
Radial clearance of piston	μm	10

An examination of the model shows that the soft switch is very sensitive to operating parameters. The radial clearance, maximum displacement, spring preload, spring rate, and unlocking distance of the piston all need to be tuned to ensure that the soft switch behaves correctly. It was determined that in order to validate the function of unlocking mechanism, it would be necessary to make as many of the soft switch parameters variable as possible. This necessitated an increase in size of the soft switch from the optimized values. The model was used to ensure that as the scale of the soft switch components increased, functionality would remain. Table 2 includes the new dimensions and operation parameters of the larger soft switch components.

To facilitate an understanding of how the soft switch components interact, the soft switch can be seen in its fully assembled state in Fig. 4. The piston is manufactured out of austenitic stainless steel. A magnet is embedded on the left side of the piston along the centerline, to generate a magnetic field that is used to determine piston position. There is a large bore on the right side of the piston large enough for the compression spring to reach the bottom and seat the piston. There are a number of shims seated under the right side of the compression spring to change the precompressive force on the piston. The set screw on the right adjusts the maximum displacement of the piston by providing a physical stop to its movement. The piston travels inside a bronze sleeve, the position of which is adjustable with shims to change the unlocking distance of the piston. The sleeve is seated inside an aluminum body with ports on the right top and bottom that lead to the check valves, a port in the middle on the bottom to tank, and a port on the left enabling the whole aluminum block to be connected directly to the switched volume via the 3-way valve body.

Copyright © 2014 by ASME

Downloaded From: http://proceedings.asmedigitalcollection.asme.org/ on 01/29/2016 Terms of Use: http://www.asme.org/about-asme/terms-of-use

Table 2 - Dimensional and operating parameters for the numerical and physical models during unlocked soft switch testing.

			Value		
Parameter	Symbol	Units	Numerical Model	Baseline Experiment	Soft Switch
Switching period	T	s		0.15	
Duty ratio	$Duty$	%	60	50	60
Flow duty ratio	$Duty_{flow}$	%	20.0	31.1	32.2
Switched Volume	V_{switch}	cm^3		136	
Entrained air fraction at P_0	R	fraction		0.02	
Load pressure	P_{load}	MPa		5.53	
Atmospheric pressure	P_0	kPa		101	
Tank pressure	P_{tank}	kPa		101	
Mass density of oil	ρ	kg/m^3		876	
Dynamic viscosity of oil	μ	Pa*s		0.02	
Bulk modulus of air free oil	β	GPa		1.9	
Ratio of specific heats for air	γ	unitless		1.4	
3-way valve parameters:					
Discharge coeff. × load area	$C_d A_l$	mm^2		6.35	
Discharge coeff. × tank area	$C_d A_t$	mm^2		5.98	
Delay before load transition	$delay_l$	ms		41	
Delay before tank transition	$delay_t$	ms		24	
Transition time to load	tt_l	ms		12	
Transition time to tank	tt_t	ms		22	
Gap between load and tank	x_{tt}	μm		127	
Check valve parameters:					
Load cracking pressure	$P_{crk,l}$	kPa		103	
Tank cracking pressure	$P_{crk,t}$	kPa		20.7	
Load line cracking pressure	$P_{crk,HSV}$	kPa		155	
Discharge coeff. × load area	$C_d A_{ch,l}$	mm^2		4.5	
Discharge coeff. × tank area	$C_d A_{ch,t}$	mm^2		14.1	
Load line dis. coeff × area	$C_d A_{ch,HSV}$	mm^2		3.13	
Load poppet clearance	c_l	mm		1.44	
Tank poppet clearance	c_t	mm		1.71	
Soft switch parameters:					
Piston diameter	d_s	mm	12.7	NA	12.7
Max travel	$x_{s,max}$	mm	12.7	NA	12.7
Spring preload	$F_{preload}$	N	18.5	NA	18.5
Spring rate	k_s	N/m	4014	NA	4014
Distance to unlock	x_{crk}	mm	5.08	NA	5.08
Dead volume behind	V_{dead}	cm^3	1.87	NA	1.87
Radial piston clearance	c	μm	12.7	NA	12.7

Copyright © 2014 by ASME

Downloaded From: http://proceedings.asmedigitalcollection.asme.org/ on 01/29/2016 Terms of Use: http://www.asme.org/about-asme/terms-of-use

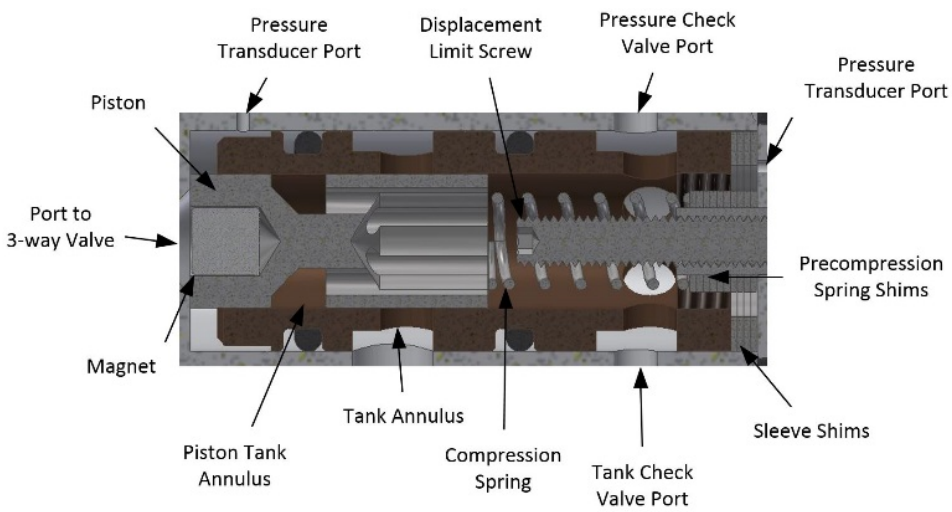

Figure 4 - Soft switch assembly.

An image of the test circuit with component labels can be seen in Fig. 5. Component part numbers and specifications can be cross-referenced in Table 3. The variable displacement axial piston pump has a pressure compensator on it that has been defeated so that for these tests the pump is behaving as a fixed displacement source. Notice that the switched volume is minimum by mounting the 3-way valve and soft switch directly to the pump. This minimized switched volume decreases the magnitude of the compressibility losses within the system. For this same reason the check valves from/to the back of the soft switch are mounted in the soft switch manifold to decrease dead volume behind the piston. The 3-way valve is controlled via a transistor circuit (not pictured) emitting 24 volts to the valve solenoid at the duty ratio given in Table 2.

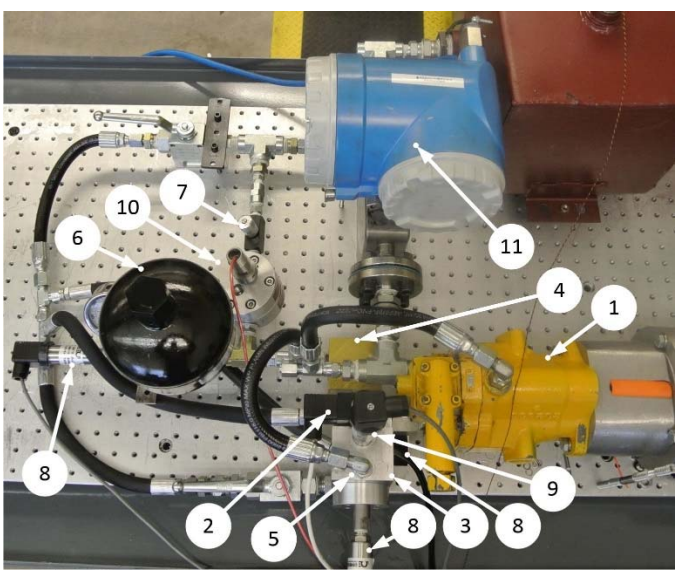

Figure 5 - Soft switch test circuit.

Table 3 - Components in the virtually variable displacement pumping circuit.

#	Name	Part	Specs.
1	Var. Disp. Pump		3+ gpm
2	3-way Valve	Hydroforce SV08-30	20.7 MPa
3	Soft Switch		
4	Load line check valve	Eaton CV6-10	76L/min 35 MPa
5a	Load check valve	Inserta ICS-C-04	3 psi crack 5 GPM nom.
5b	Tank check valve	Inserta ICS-C-02	15 psi crack 2 GPM nom.
6	Accumulator	Parker Diaphragm	20 MPa 1 L
7	Needle Valve		
8	Position Sensor	Honeywell HMC1512	120 mV
9	Pressure Transducers	Omega PX4201	20.7 MPa 100 mV
10	Gear Flow Meter	AW Gear Meters JVA-20KG-25	.01-2 gpm 5000 psi
11	Coriolis Flow Meter	Endress+Hauser Promass 80E08	
11	Thermocouple Amp.	AEM 302204	K-Type 1000 C max

In order to determine if the pressure signal controlled locking mechanism operates as expected, two test cases were run. The first test case uses the pulse width modulated pumping circuit shown in Fig. 2. The second test case is a control case that utilizes the same hydraulic circuit with the soft switch in place but with the piston completely locked at zero displacement.

Copyright © 2014 by ASME

Downloaded From: http://proceedings.asmedigitalcollection.asme.org/ on 01/29/2016 Terms of Use: http://www.asme.org/about-asme/terms-of-use

If the soft switch is unlocking correctly, a small portion of the flow that normally goes to the load in a PWM system without a soft switch will be absorbed by the soft switch and then flow to tank. This effectively reduces the duty ratio of the circuit. In order to compare the efficiencies of the two test cases, their flow duty ratios, which are defined as the flow rate reaching the load divided by the flow rate of the pump, need to be the same. This necessitates a decrease in the valve duty ratio for the control case. A summary of the test parameters for both cases is viewable in Table 2.

RESULTS

This section will describe a prediction of the soft switch operation in the described virtually variable displacement circuit. These predictions will be compared to the results of the two test cases described in the previous section.

The parameters listed in Table 2 for the numerical model with soft switch unlocked were used in the model developed in Section 3. The model was run for multiple cycles to reach cyclic steady-state and then compared to the physical system. The model predicts the piston displacement seen in Fig. 6. At 0.19 seconds the piston begins to displace as a result of the 3-way valve transitioning from load to tank. The switched volume pressure continues to rise until the forces on the front and back of the soft switch balance each other. Notice that the piston continues to displace, and then as the 3-way valve transitions back to tank the piston reaches the unlocking distance and quickly absorbs the flow. It is then reseated in preparation for the next switching cycle.

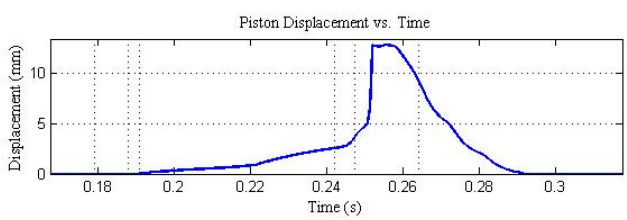

Figure 6 - Piston displacement with respect to time for the model simulation.

When the experimental system is run with the same parameters as the simulation, a similar piston trajectory is traced. Fig. 7 overlays the results of the physical test with the simulation results. The first portion of the piston displacement is similar, but as the piston reaches the unlocking distance the profiles differ. The simulation predicts that once the piston is unlocked displacement will occur much faster than is seen in the physical system. It is interesting to note that the reverse is true for piston reseating, with the physical system reseating at a much faster rate than predicted.

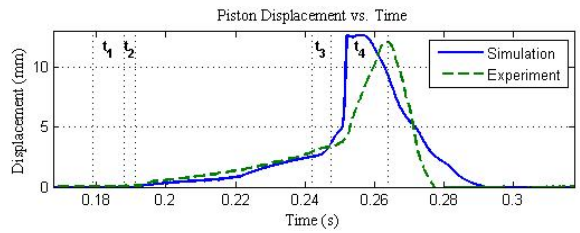

Figure 7 - Piston displacement with respect to time for the model and simulation together.

Fig. 8 compares the piston trajectory and switched volume pressure as a function of time for the physical control and soft switch test cases. Notice that the command duty ratios are different between the two cases in order for them to have similar flow duty ratios. In both cases the switched volume pressure builds to approximately the same value at the same rate. Notice the large pressure spike in the control case, which is a result of the 3-way valve transitioning back to tank. During this transition, fluid is being throttled across the 3-way valve with a high pressure differential, while in the soft switch case that pressure spike is completely eliminated.

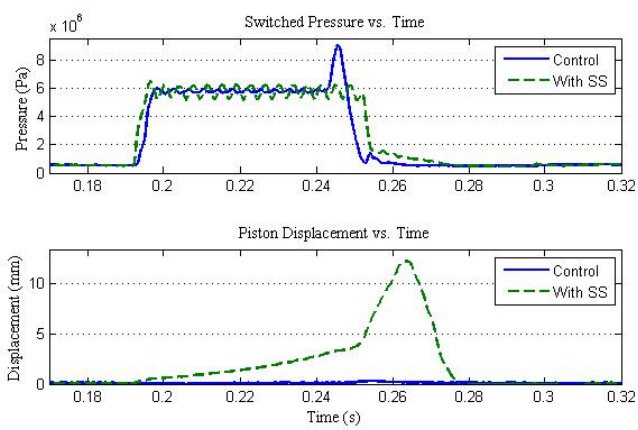

Figure 8 - Switched volume pressure and piston displacement for the control and soft switch experiments.

The efficiency of each virtually variable displacement circuit test/simulation can be calculated by taking the ratio of the energy going to load per cycle (Eqn. 15) to the energy provided to the system by the pump per cycle (Eqn. 16).

$$E_{load} = \int_0^T (P_{load} - P_{tank}) Q_{load}\, dt \qquad (15)$$

$$E_{load} = \int_0^T (P_{switched} - P_{tank}) Q_{pump}\, dt \qquad (16)$$

The energy analysis shown above shows that the circuit with the soft switch locked is 66.3% efficient at a flow duty ratio of 31.1%, while the soft switch circuit is 64.9% efficient at a flow duty ratio of 32.2%.

Copyright © 2014 by ASME

Downloaded From: http://proceedings.asmedigitalcollection.asme.org/ on 01/29/2016 Terms of Use: http://www.asme.org/about-asme/terms-of-use

DISCUSSION

The results illustrate that the soft switch lock-release mechanism functions as designed, absorbing flow during the pressure to tank 3-way valve transition. In Fig. 8, it can be seen that the piston displacement increases gradually during the on-state due to leakage and then rapidly absorbs unlocks and absorbs flow during the valve transition. This eliminates the pressure spike and throttling loss that would otherwise occur (as seen in the control case).

Unfortunately it appears that while operating under the test parameters used in these experiments that system efficiency is not being increased. Looking at the previous optimization and energy loss analysis completed by Van de Ven [7] these results are not un-expected. It is shown in that work that for the circuit setup used in this proof of concept, the efficiency of the system compared to the check valve control case is not very different. However, when using both passive and active soft switches, the potential energy savings is 66% over a baseline relief valve circuit [7]. Another point to remember is that in prior analysis the soft switch was operating under optimized parameters, which is not the case in this work.

There are a few sources of measurement error associated with these trials that should be noted. When measuring the flow rate of the pump it would be ideal to gather flow information from downstream. In this case the Coriolis meter was placed between the tank and the pump in an effort to keep the switched volume of the circuit as low as possible, which allows for better soft switch behavior. There is some uncertainty about the 3-way valve spool position during the switching cycles since there is a delay in the solenoid receiving the transition signal and it overcoming the inductance of the coil enough to displace the spool. Future work will resolve that uncertainty by creating a valve position profile, which in turn will allow for more accurate model predictions of the valve transitions.

The gear flow meter measured the load flow rate, but the gears have enough inertia that small flow pulses were not detectable. This error was mitigated by the accumulator on the load line which results in the load line being the location with the most stable flow.

Perhaps the largest and most difficult measurement errors to handle are associated with the position sensing of the soft switch. The position sensor operates by sensing the angle of any nearby magnetic fields. A magnet was embedded in the piston, which was made of austenitic stainless steel, which is almost completely nonmagnetic. Most of the other components surrounding the piston were made of materials that are nonmagnetic, but the proximity of the solenoid for the 3-way valve presented a significant challenge. It was found that the output of the sensor could be calibrated to remove the predictable effects of the magnetic field of the solenoid increasing and then collapsing. Each new position of the sleeve and change in duty ratio or switching period of the valve requires a separate calibration. In the end the position of the piston is known with only relative certainty.

CONCLUSIONS

The soft switch locking mechanism behaves as expected. The model presented in this paper does a fair job of predicting the function of the circuit under different parameters. However, more work is required in order to bring the model and experiments into better agreement.

Though the locking mechanism operates, the parameters in this study do not allow it to operate in a manner that increases the efficiency of the system compared to a normal PWM circuit. The design of the soft switch and model together will give more understanding about which parameters most affect operation and therefore which parameters need to be focused on in further optimization and design.

REFERENCES

[1] Li, P. Y., Li, C. Y., and Chase, T. R., 2005, “Software Enabled Variable Displacement Pumps,” ASME International Mechanical Engineering Congress and Exposition, Orlando, FL, November 5–11, ASME Paper No. IMECE2005-81376, pp. 63–72.

[2] Mohan, N., Robbins, W. P., and Undeland, T. M., 1995, Power Electronics: Converters, Applications and Design, Wiley and Sons, New York.

[3] Cao, J. Gu, L., Wang, F., and Qiu, M., 2005, “Switchmode Hydraulic Power Supply Theory,” ASME International Mechanical Engineering Congress and Exposition, Orlando, FL, November 5–11, ASME Paper No. IMECE2005-79019, pp. 85–91.

[4] Tu, H. C., Rannow, M. B., Wang, M., Li, P. Y., Chase, T. R., and Van de Ven, J. D., 2012, “Design, Modeling and Validation of a High-Speed Rotary Pulse-Width-Modulation On/Off Hydraulic Valve,” ASME J. Dyn. Syst., Meas. Control, 134(6), p. 061002.

[5] Wu, J., and Van de Ven, J. D., 2010, “Development of a High-Speed On-Off Valve for Switch-Mode Control of Hydraulic Circuits With Four-Quadrant Control,” ASME International Mechanical Engineering Congress & Exposition, Vancouver, BC, Canada, November 12–18, ASME Paper No. IMECE2010-37380, pp. 281–288.

[6] Rannow, M. B., and Li, P. Y., 2012, “Soft Switching Approach to Reducing Transition Losses in On/Off Hydraulic Valve,” ASME J. Dyn. Syst., Meas. Control, 134(6), p. 064501.

[7] Van de Ven, J.D., 2014, “Soft Switch Lock-Release Mechanism for a Switch-Mode Hydraulic Pump Circuit,” ASME J. Dyn. Syst., Meas. Control, 136(3), p. 031003.

[8] Cho, B.-H., Lee, H.-W., and Oh, J.-S., 2002, “Estimation Technique of Air Content in Automatic Transmission Fluid by Measuring Effective Bulk Modulus,” Int. J. Autom. Technol., 3(2), pp. 57–61.

[9] Çengel, Y. A., and Cimbala, J. M., 2010, Fluid Mechanics: Fundamentals and Applications, McGraw-Hill, New York.

Copyright © 2014 by ASME

Downloaded From: http://proceedings.asmedigitalcollection.asme.org/ on 01/29/2016 Terms of Use: http://www.asme.org/about-asme/terms-of-use